

Task 6.1P

Evidence of Learning

Exercise 4.2

Consider we have a network with a subnet along with prefix 192.168.40.128/26. Can you list down two IP addresses that can be assigned to two PCs connected to this network?

The /26 CIDR prefix means that 26 bits are reserved for the network and the remaining $32-26=6$ bits are for the host. This means that there can be $2^6 = 64$ total addresses, out of which, we will have $64-2=62$ usable addresses. The reason we removed two from here is because one is the network address and the other one is for broadcast.

From $256-64 = 192$, the /26 subnet mask is 255.255.255.192. Therefore, the subnet ranges for the last octet are 0-63, 64-127, 128-191 and 192-255. Because the network address is 192.168.40.128, we will be dealing with the 128-191 set. This means the first usable IP is 192.168.40.129, the last usable IP is 192.168.40.190 and the broadcast address is 192.168.40.191.

This means that the two PCs can use addresses like: 192.168.40.129 and 192.168.40.130.

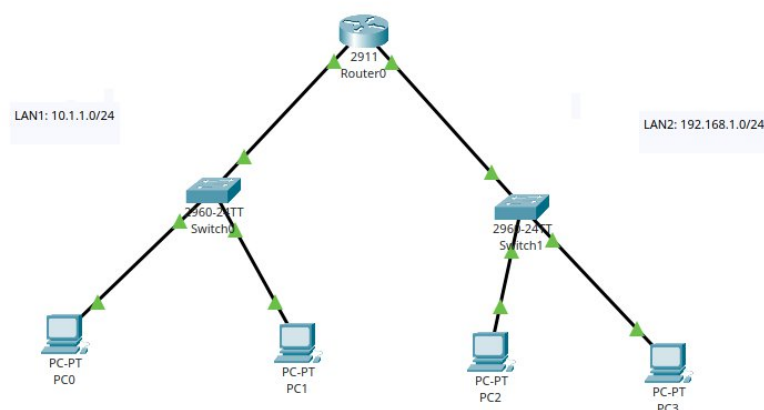
Exercise 4.3.1

Open the installed Cisco Packet Tracer and get yourself familiarised with CiscoPacket Tracer. Done.

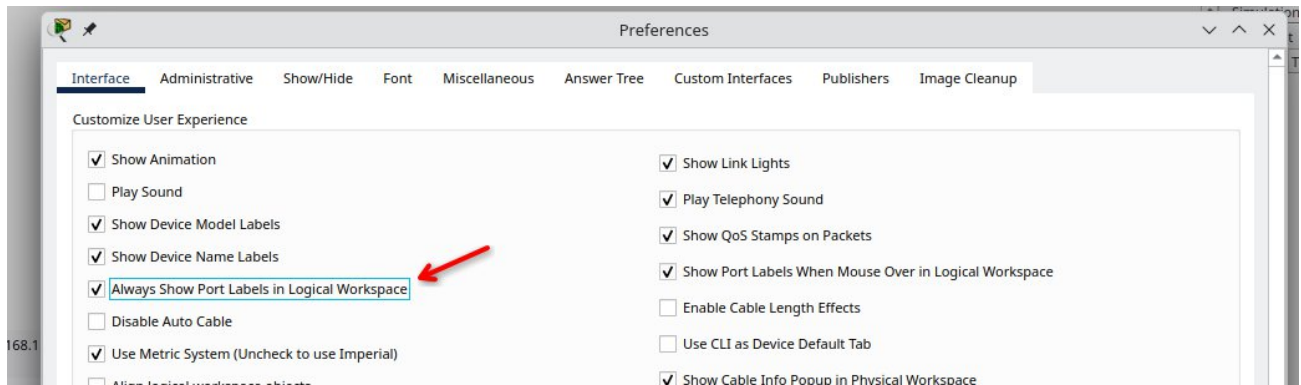
Exercise 4.3.2

When you are working in Cisco Packet Tracer, can you show all the port labels in your logical workspace? Tip: Check Options > Preferences

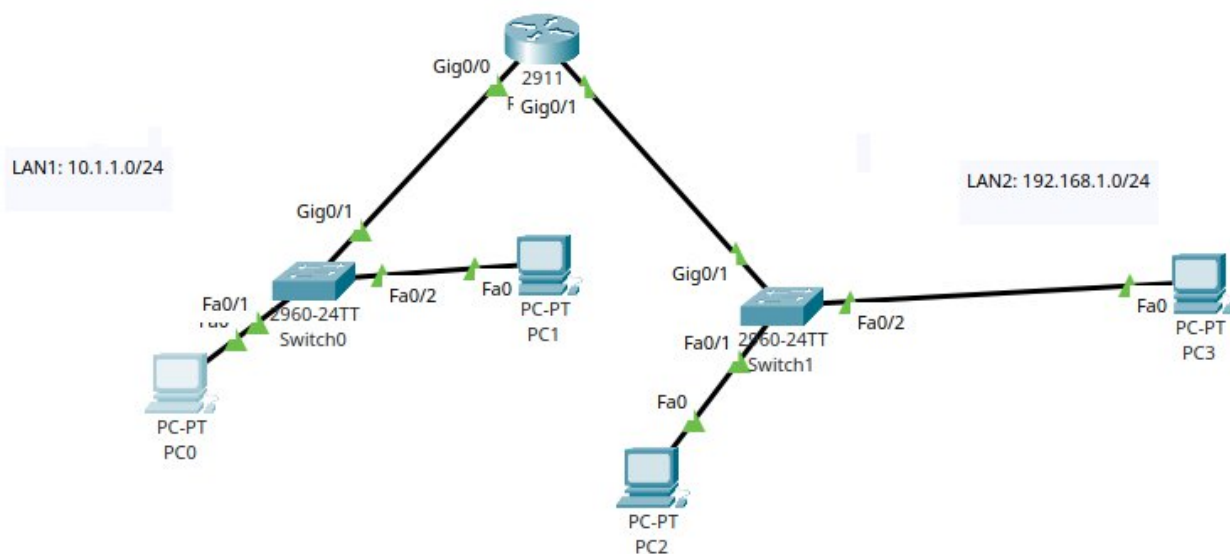
This is what it looks like now, without this option turned on.



I then checked this option in Option —> Preferences.



Now, this is what it looks like. I had to rearrange the items here to ensure that all the labels are clearly visible and readable.

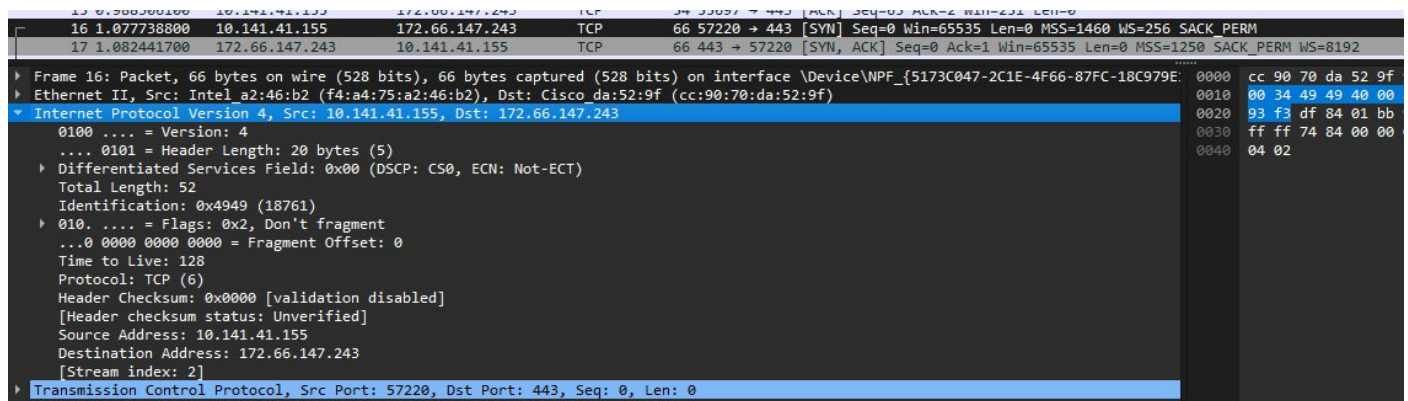


Active Class Week 6: 6.2 Activity 1

I used two packets that were outgoing from my capture. The first one is frame 16, which is a TCP SYN packet sent to port 443. The second one is frame 7, which is a DNS request sent using UDP.

1. Examine the IP datagram's header. Can you identify the IP address of your device and the IP address of the destination host?

For the first analysis, I selected frame 16. The IP address of my device is 10.141.41.155. The destination host IP address is 172.66.147.243. I identified them using the Source Address and Destination Address fields inside the Internet Protocol Version 4 section.



2. Can you identify the version of IP addresses?

As seen in the above image, the Version field is set to 4. So, both addresses are IPv4 addresses. They are also written using the dotted-decimal IPv4 format.

3. By examine the IP header, can you identify the transport layer protocol used? Does this match with the transport layer protocol listed in in the packet details window?

The Protocol field in the IP header shows TCP (6). Protocol number 6 represents Transmission Control Protocol. This matches with the Transmission Control Protocol section shown separately in the packet details window. The packet uses source port 57220 and destination port 443. It is a SYN packet, so it's safe to assume that it's starting a connection to an HTTPS service.

4. What is the size of IP header in Bytes? What is the size of the payload (in Bytes) of this IP datagram? How did you calculate the size of the payload?

The Header Length field shows the IP header is 20 bytes. The Total Length of the IP datagram is 52

IP payload = Total Length - IP header length $\rightarrow 52 - 20 = 32$ bytes

Therefore, the IP payload is 32 bytes. This contains the TCP header and its options. Since this is a SYN packet, it doesn't contain any application data yet. The captured frame size is 66 bytes, but I didn't use that value because it also includes the 14-byte Ethernet header.

5. Can you check whether this IP datagram is fragmented or not? Explain your answer.

This IP datagram is not fragmented. The screenshot shows the Flags field as 0x2, with Wireshark labelling it as Don't fragment. This means the Don't Fragment flag is set.

The Fragment Offset is also 0, and the More Fragments flag is not set. Together, these fields confirm that the packet is a complete IP datagram and not one part of a fragmented packet.

6. Examine other important fields in the IP header.

The screenshot shows several other important fields:

- Differentiated Services Field: The value is 0x00, with DSCP set to CS0. This means the packet is using the default service level without any special traffic priority.
- ECN: It shows Not-ECT, meaning the packet is not using Explicit Congestion Notification.
- Total Length: The complete IP datagram is 52 bytes, including the 20-byte IP header and 32-byte payload.
- Identification: The value is 0x4949, or 18761. This value can be used to identify and reassemble fragments if fragmentation happens.
- Time to Live: The TTL is 128. Each router reduces this value by one, and the packet is discarded if it reaches zero.
- Protocol: The value is TCP (6), showing that the IP payload contains a TCP segment.
- Header Checksum: The value is 0x0000, with validation disabled and the status shown as unverified. This means Wireshark did not verify the checksum, so it does not automatically mean the header is invalid.
- Source Address: 10.141.41.155
- Destination Address: 172.66.147.243

7. Now, conduct a similar analysis using a different type of packet (if you have used HTTP before, now you can use DNS) and compare the findings.

```
7 0.364651700 10.141.41.155 10.128.164.31 DNS 71 Standard query 0x7fde AAAA example.com
8 0.382303200 10.128.164.31 10.141.41.155 DNS 137 Standard query response 0x7fde AAAA example.com AAAA 3605/1700/10/1/33/1/1705 AAAA 3605/1700/10/1/33/1/1705

Frame 7: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF_{5173C047-2C1E-4F66-87FC-18C979E10B8D}, id 0
  Ethernet II, Src: Intel_a2:46:b2 (f4:a4:75:a2:46:b2), Dst: Cisco_da:52:9f (cc:90:70:da:52:9f)
    Destination: Cisco_da:52:9f (cc:90:70:da:52:9f)
    Source: Intel_a2:46:b2 (f4:a4:75:a2:46:b2)
    Type: IPv4 (0x0800)
    [Stream index: 0]
  Internet Protocol Version 4, Src: 10.141.41.155, Dst: 10.128.164.31
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ▾ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 57
    Identification: 0x218a (8586)
    ▾ 0000 .... = Flags: 0x00
      0... .... = Reserved bit: Not set
      .0.. .... = Don't fragment: Not set
      ..0. .... = More fragments: Not set
      ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.141.41.155
    Destination Address: 10.128.164.31
    [Stream index: 1]
  User Datagram Protocol, Src Port: 52614, Dst Port: 53
    Source Port: 52614
    Destination Port: 53
    Length: 37
    Checksum: 0xe2fd [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
    ▸ [Timestamps]
    UDP payload (29 bytes)
  Domain Name System (query)
    Transaction ID: 0x7fde
    ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    ▸ Queries
    [Response In: 8]
```

For the second analysis, I selected frame 7. This is a DNS query for example.com, sent from my device using UDP.

The Source Address is 10.141.41.155, which is still my device. The Destination Address is 10.128.164.31, which is the DNS server.

The Version field is 4, so this is also an IPv4 datagram. The DNS query is asking for an AAAA record, but the packet carrying the request is still IPv4.

The Protocol field shows UDP (17). This matches with the User Datagram Protocol section in the packet details. The source port is 52614 and destination port is 53, which is the normal DNS port.

The IP header length is 20 bytes and the Total Length is 57 bytes.

IP payload = 57 – 20 = 37 bytes.

The 37-byte payload includes an 8-byte UDP header and a 29-byte DNS message.

This packet is also not fragmented. Its More Fragments bit is 0 and Fragment Offset is 0. The Don't Fragment bit is 0, which means fragmentation was allowed, but the packet itself still wasn't fragmented.

Other fields in this packet include:

- DSCP: 0
- ECN: 0
- Identification: 0x218a or 8586
- Time to Live: 128
- Protocol: UDP (17)
- Header Checksum: 0x0000

Both packets use IPv4, have a 20-byte IP header, uses the same source address and has a TTL of 128. The main difference is frame 16 uses TCP and frame 7 uses UDP. The TCP packet has a 32-byte IP payload, while the DNS/UDP packet has a 37-byte IP payload. Frame 16 also has the Don't Fragment bit set, while frame 7 does not.

Active Class Week 6: 6.3 Activity 2

1. Assume you are sending a present to a friend in another country. Can you list the various places and steps that your parcel would go in the postal system before it reaches your friend?

The parcel would go through these places and steps:

1. I pack the present and write my friend's address and my return address.
2. I take it to a local post office or arrange for it to be collected.
3. The parcel goes to a local sorting centre.
4. It is sorted based on the destination country and sent to an international sorting centre.
5. The parcel is transported by air or sea to the destination country.
6. It goes through customs and security checks.
7. The destination country's postal service receives and sorts it.
8. It is sent to the local delivery centre closest to my friend.
9. A delivery driver takes the parcel to my friend's address.

Each postal centre only needs to decide where the parcel should go next. It doesn't need to know every part of the complete journey.

2. How this analogous to a situation where you want to send a message to a friend in another country over the computer networks?

Sending a message over a computer network works in a similar way. The message is divided into packets before being sent. Each packet contains a source IP address and destination IP address, similar to the sender and receiver addresses written on a parcel.

The packet first goes through the local network to the default gateway. Routers then examine the destination IP address and use their routing tables to decide the next place the packet should go. This continues until it reaches the destination network and the receiver's device.

The postal and network parts can be compared like this:

Postal system	Computer network
Present or parcel	Data packet
Sender address	Source IP address
Friend's address	Destination IP address
Local post office	Default gateway
Sorting centre	Router
Road, air or sea transport	Ethernet, fibre or wireless link
Postal route information	Routing table
Local delivery centre	Destination router
Friend receiving the parcel	Destination device receiving the packet

A message may contain several packets. These packets can sometimes take different routes, but they are placed in the correct order when needed at the destination after they are recieved.

3. Assume we need to build the following network with two LANs (LAN1 and LAN2). Each group member has a role to play. One group member can be the router and four other group members could be PCs (PC1 and PC2 belong to LAN1 and PC3 and PC4 belong to LAN2). Each device needs to set their own network configuration. The Router needs to set its interfaces/port and PCs need to set its IP address and gateways to be able to make a communication between two LANs. Discuss the configurations of your own device with your group members.

The diagram contains two separate LANs:

1. LAN1 uses the 10.1.1.0/24 network.
2. LAN2 uses the 192.168.1.0/24 network.

A /24 network uses the subnet mask 255.255.255.0. Each device needs a unique IP address within its own network. The router interface connected to each LAN will be used as the default gateway. A possible example configuration is:

Device	IP address	Subnet mask	Default gateway
Router interface for LAN1	10.1.1.1	255.255.255.0	Not required
PC1	10.1.1.2	255.255.255.0	10.1.1.1
PC2	10.1.1.3	255.255.255.0	10.1.1.1
PC3	192.168.1.2	255.255.255.0	192.168.1.1
PC4	192.168.1.3	255.255.255.0	192.168.1.1

The router member can configure its two interfaces using commands similar to these:

```
enable
configure terminal

interface GigabitEthernet0/0
ip address 10.1.1.1 255.255.255.0
no shutdown
exit

interface GigabitEthernet0/1
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

end
```

The exact interface names may change depending on the router model. Static routes aren't needed here because both networks are directly connected to the same router.

The switches don't need an IP address to forward normal Ethernet frames. They would only need an IP address if remote switch management is required.

4. Assume PC1 needs to send a packet to PC3, discuss the steps that the packet needs to go through to reach to PC3.

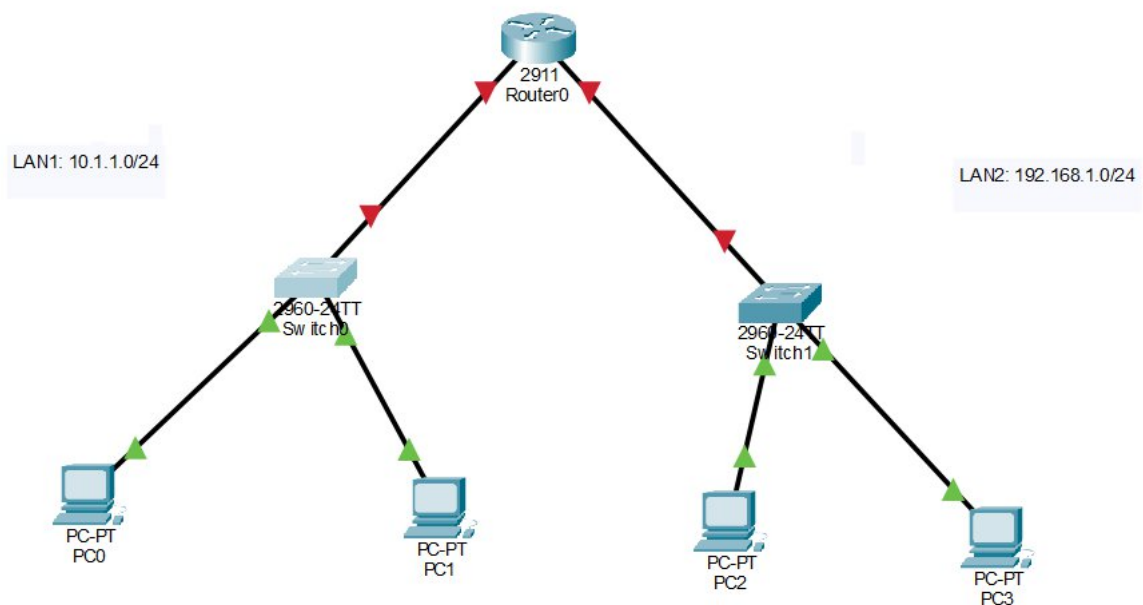
The steps will be as follows:

1. PC1 creates a packet with 10.1.1.2 as the source IP address and 192.168.1.2 as the destination IP address.
2. PC1 compares the destination address with its own network. It can see that PC3 belongs to a different network, so the packet must be sent to the default gateway at 10.1.1.1.
3. PC1 uses Address Resolution Protocol, or ARP, to find the MAC address of the router interface if it isn't already stored in its ARP table.
4. PC1 places the packet inside an Ethernet frame. The frame uses PC1's MAC address as the source and the router's LAN1 MAC address as the destination.
5. Switch 1 receives the frame and forwards it to the router.
6. The router removes the Ethernet frame and examines the destination IP address. Its routing table shows that 192.168.1.0/24 is directly connected through the interface leading to LAN2.
7. The router uses ARP to find PC3's MAC address if needed. It then places the same IP packet inside a new Ethernet frame. This frame uses the router's LAN2 MAC address as the source and PC3's MAC address as the destination.
8. Switch 2 forwards the frame to PC3. PC3 removes the Ethernet frame, checks the destination IP address, and passes the received data to the correct application.

The source and destination IP addresses stays as 10.1.1.2 and 192.168.1.2 through the full process. But the source and destination MAC addresses changes when the packet goes through the router.

Active Class Week 6: 6.3 Activity 2

I added all the devices to packettracer and connected them with Copper Straight Through cables.

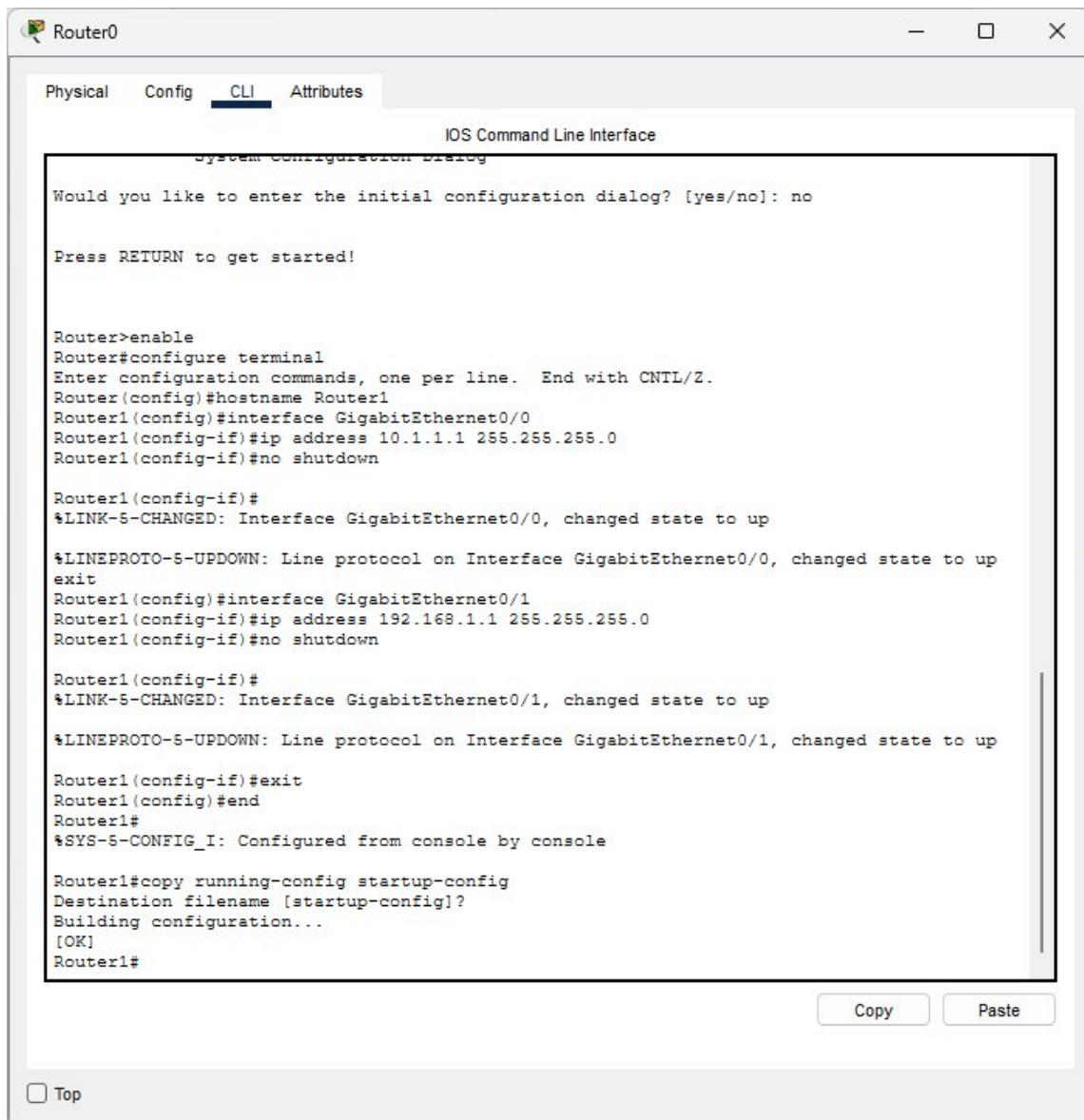


I then started to configure the router.

```
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>
```



After that, I verified that everything has been set as required for the network to function in the given configuration.

```
Router1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.1.1.1        YES manual up          up
GigabitEthernet0/1       192.168.1.1     YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router1#
```

```
Router1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

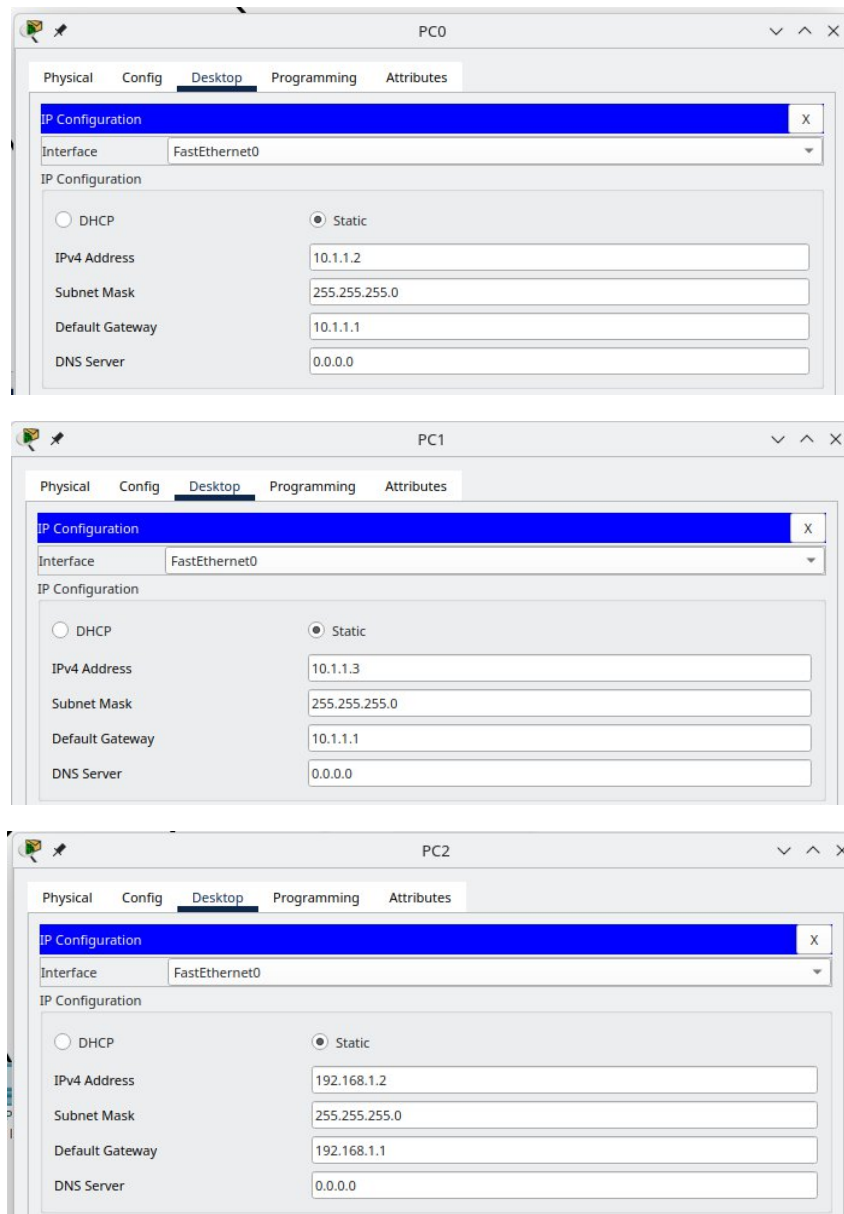
Gateway of last resort is not set

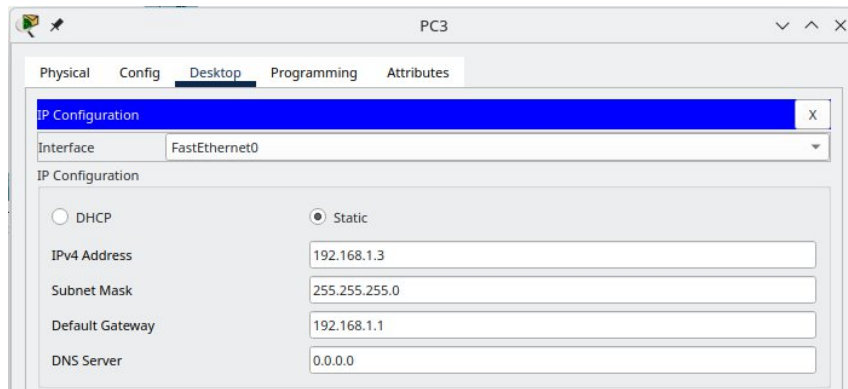
```

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/1
L    192.168.1.1/32 is directly connected, GigabitEthernet0/1

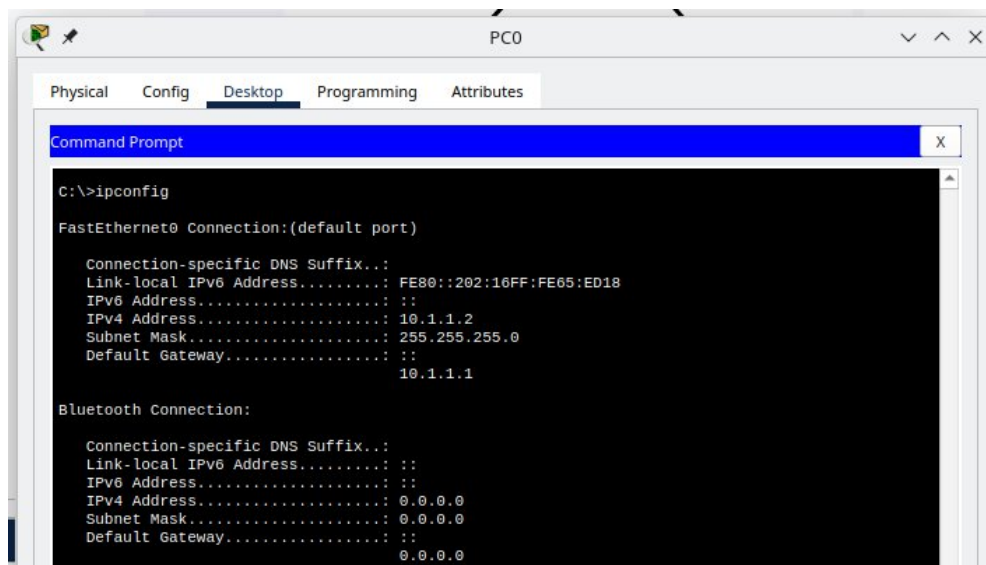
```

I then assigned static IP addresses for all the computers. PC0 and PC1 are in LAN1 and PC2 and PC3 are in LAN2.

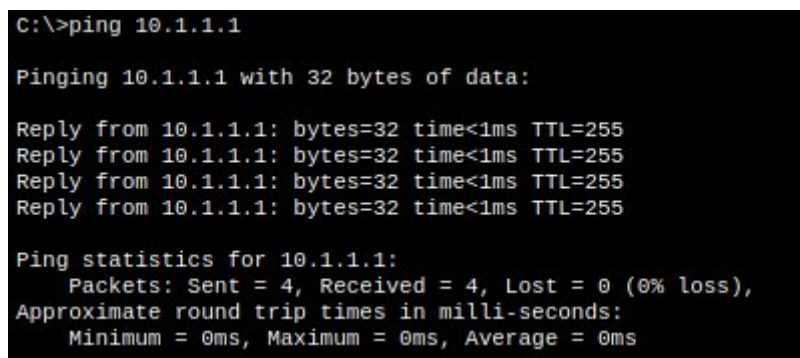




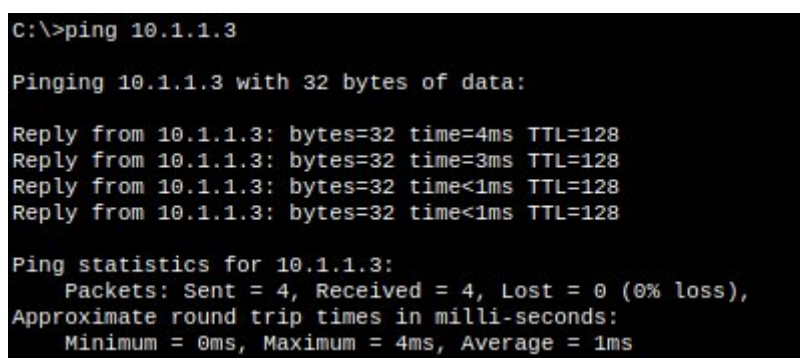
I then opened PC0's Command Prompt to test connectivity. I initially configured the IP configuration before proceeding to ping the devices in the network.



I can ping the router.



I can ping PC1, which is in the same network / LAN1.



I can also ping the other router interface.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Along with PC2 in LAN2

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

And PC3 in LAN2.

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

When pinging devices in the other LAN, the first ICMP Echo Request timed out in both tests. This happened because ARP was still finding the MAC addresses needed to forward the packet. When I ran the ping again, it didn't time out because the required addresses were already saved in the ARP tables.

Demonstration of the Simulation: <https://youtu.be/76peT8rW1XM>

Simulation of a single ping showing the ICMP Echo Request and Echo Reply between PC0 and PC3:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
Visible	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Router0	ICMP
	0.003	Router0	Switch1	ICMP
	0.004	Switch1	PC3	ICMP
	0.005	PC3	Switch1	ICMP
	0.006	Switch1	Router0	ICMP
	0.007	Router0	Switch0	ICMP
	0.008	Switch0	PC0	ICMP

PDU Information at Device: Router0

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Router0

Source: PC0

Destination: 192.168.1.3

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3: IP Header Src. IP: 10.1.1.2. Dest. IP: 192.168.1.3 ICMP Message Type: 8

Layer2: Ethernet II Header 0002.1665.ED18 >> 0030.A379.0301

Layer1: Port GigabitEthernet0/0

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3: IP Header Src. IP: 10.1.1.2. Dest. IP: 192.168.1.3 ICMP Message Type: 8

Layer2: Ethernet II Header 0030.A379.0302 >> 0001.97E5.026D

Layer1: Port(s): GigabitEthernet0/1

1. The CEF table has an entry for the destination IP address.

2. The device decrements the TTL on the packet.

The OSI Model shows Router0 receiving the ICMP Echo Request through GigabitEthernet0/0 and forwarding it through GigabitEthernet0/1. The source IP stays as 10.1.1.2 and the destination IP stays as 192.168.1.3. Router0 checks its CEF table for the destination and decreases the packet's TTL before forwarding it. The Ethernet MAC addresses change because the router creates a new frame for LAN2.